

Answer Key — Cells and Batteries

Final answer with a one-line reason. Full methods are in the Notes' worked examples.

Objective

- Q1 **(b) chemical reactions inside it** — a chemical reaction inside the cell makes the current.
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- Q2 **(b) electrolyte** — the electrolyte (weak acid/salt solution) conducts and reacts.
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- Q3 **(b) negative terminal** — the zinc container is the negative terminal; the carbon rod is positive.
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- Q4 **(b) lemon juice** — lemon juice is a good electrolyte; pure water is not.
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- Q5 **(b) False** — a dry cell is single-use; only rechargeable batteries can be recharged.
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- Q6 **electrodes** — the two rods are the electrodes.
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- Q7 **lithium** — lithium-ion batteries are the most common rechargeables.
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Short answer

- Q8 Two electrodes (different metals) dipped in an electrolyte (weak acid/salt solution) in a container; a chemical reaction between them produces electricity.
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- Q9 Its electrolyte is a moist paste, not a liquid. Positive = central carbon rod (with metal cap); negative = zinc container.
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- Q10 They can be recharged and reused many times, saving money and reducing waste.
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Long / application

- Q11 Insert a copper wire and an iron nail (kept apart) into each lemon; join the nail of one lemon to the copper of the next; connect the free copper and nail to the LED (longer leg to positive).
Electrodes: copper wire and iron nail; **electrolyte:** lemon juice. The chemical reaction lights the LED (reverse LED leads if it doesn't glow).
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- Q12 A "dead" cell has used up its chemicals and can no longer supply current. Rechargeables are preferred because they can be charged and reused many times, saving money and waste. Old batteries contain acids/metals that can harm the environment or catch fire, and hold valuable recyclable materials — so dispose of them at e-waste facilities.
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